



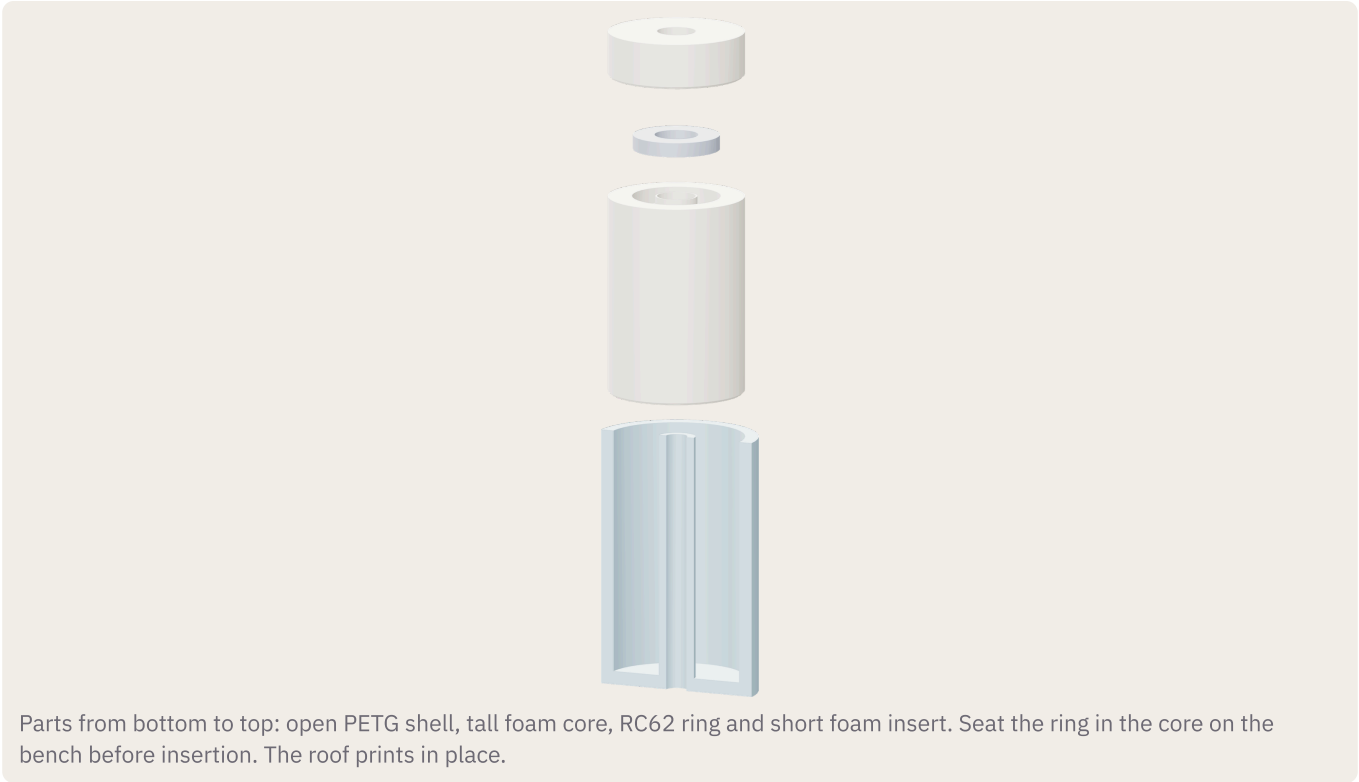
Magnetic float

Dry the filament. Print the foam. Seat the magnet. Build the PETG shell around all three pieces.



Three prints. One magnet.

A 36 mm × 60.06 mm float with a continuous PETG envelope. The printer closes the roof after the foam and magnet go in.



Parts from bottom to top: open PETG shell, tall foam core, RC62 ring and short foam insert. Seat the ring in the core on the bench before insertion. The roof prints in place.

Tall white core	ASA Aero, plate 1. Pocket faces upward.
Short white insert	ASA Aero, plate 2. Chamfered edges face downward.
One RC62 ring	K&J, nickel plated; 19.05 × 9.525 × 3.175 mm.
Clear shell	Bambu PETG Translucent Clear 32101, one plate with a programmed insertion pause.

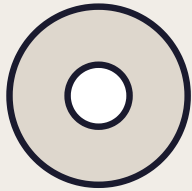
● Clear PETG, shaded blue ● Aero foam ● Part being added

BUILD ORDER

Dry overnight. Print and prepare both Aero pieces. Seat the magnet on the bench. Print PETG, insert the prepared pieces at the pause, and resume. Allow about **4 h 16 min of printing**, plus heating, cooling and insertion.

Start both dryers tonight

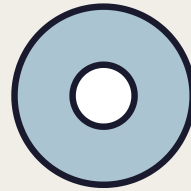
Use the two different AMS units already in the shop. The cycles run at the same time; the longer PETG cycle sets the overnight schedule.



WHITE ASA AERO

AMS HT

Right nozzle feed



CLEAR PETG

AMS 2 Pro

Left nozzle feed

ASA Aero

80 °C

8 hours

AMS HT • White 46100

PETG Translucent

65 °C

12 hours

AMS 2 Pro • Clear 32101

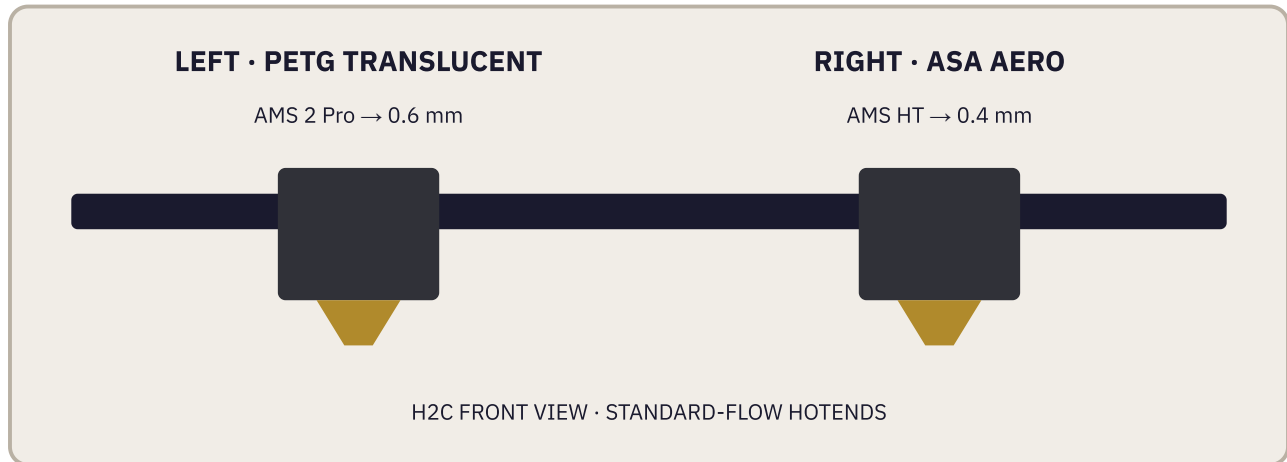
- 1 Load the ASA Aero into an **AMS HT** and the PETG Translucent into an **AMS 2 Pro**. Connect the AMS HT's own power cable. Remove PLA and other lower-temperature spools from that AMS 2 Pro.
- 2 Latch both lids. Set **static drying** to the temperatures and durations above, then start both cycles.
- 3 Complete both cycles before printing. Leave each spool in the same closed AMS for storage and printing.

KEEP THE MATERIAL NAMES EXACT

ASA Aero is the white foaming material. **Bambu PETG Translucent Clear (32101)** is the shell material, the same clear filament used for the flavor reservoirs. Select those names in the supplied projects.

Set up one H2C

The whole build uses the same nozzle pair. Foam prints on the right; the shell prints on the left.



- 1 With the printer cool, fit the **left 0.6 mm standard-flow** hotend and the **right 0.4 mm standard-flow** induction hotend.
- 2 In the printer setup, select those nozzle sizes. Route the ASA Aero AMS HT to the right feed and the PETG AMS 2 Pro to the left feed.
- 3 Set out the Engineering plate, Textured PEI plate, Bambu liquid glue, one RC62, flush cutters, a craft knife, a soft brush and a clean flat plastic scraper.
- 4 Open the supplied **3MF as a project** in Bambu Studio. Keep its printer, material and process settings when prompted; map its material to the matching AMS spool.

THE TWO PROJECTS

[magnetic-float-aero.3mf](#)

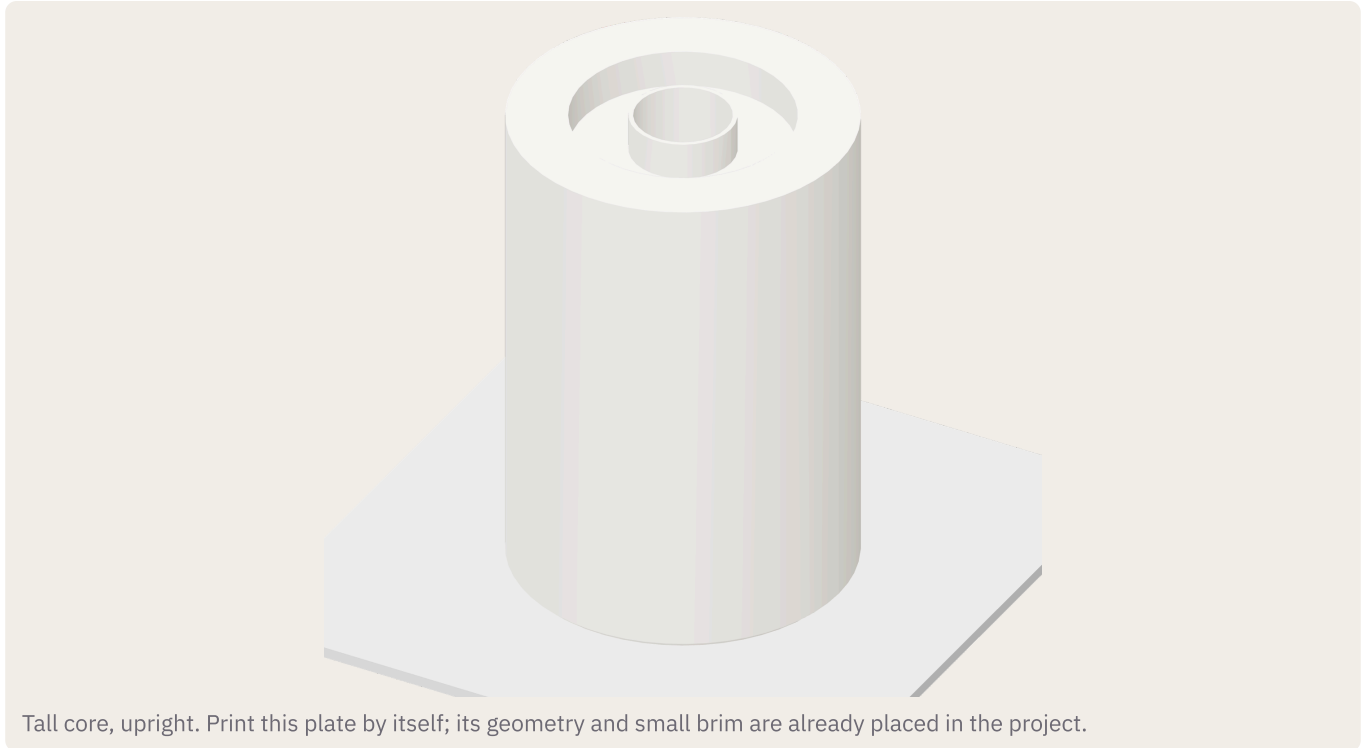
Two foam plates, printed separately.

[magnetic-float.3mf](#)

One PETG plate, with its insertion pause already included.

Print the tall core

The flat end sits on the Engineering plate. The circular magnet pocket faces upward.



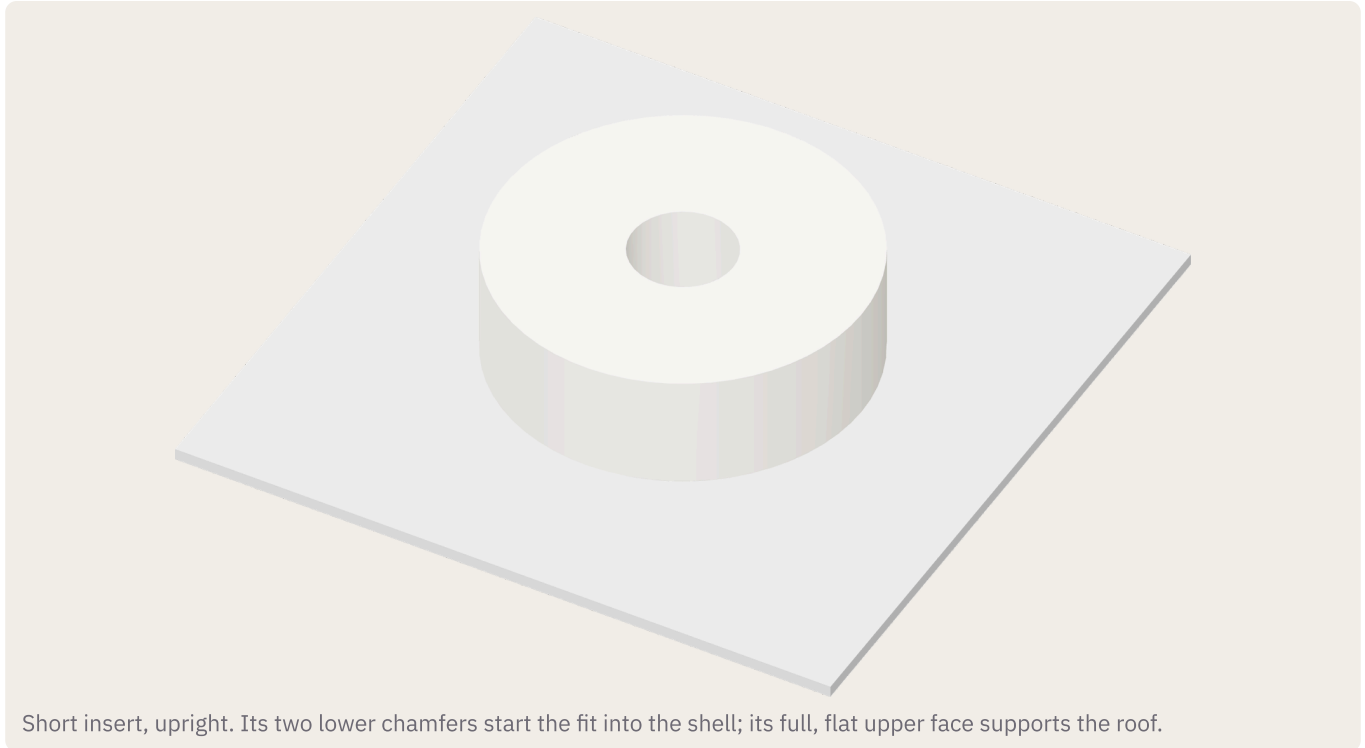
Project	magnetic-float-aero.3mf
Plate	1 · tall core
Nozzle / bed	270 / 90 °C
Chamber	60 °C
Flow / layer	0.52 / 0.20 mm

Printing: about 1 h 37 min

- 1 Wash the Engineering plate with dish detergent and warm water. Rinse and dry it.
- 2 Apply a thin, even coat of **Bambu liquid glue** over the print area and let it dry.
- 3 Select **plate 1** and the right ASA Aero feed. Close the door and lid, run room ventilation, and print.

Print the short insert

This second foam piece becomes the support directly beneath the PETG roof.

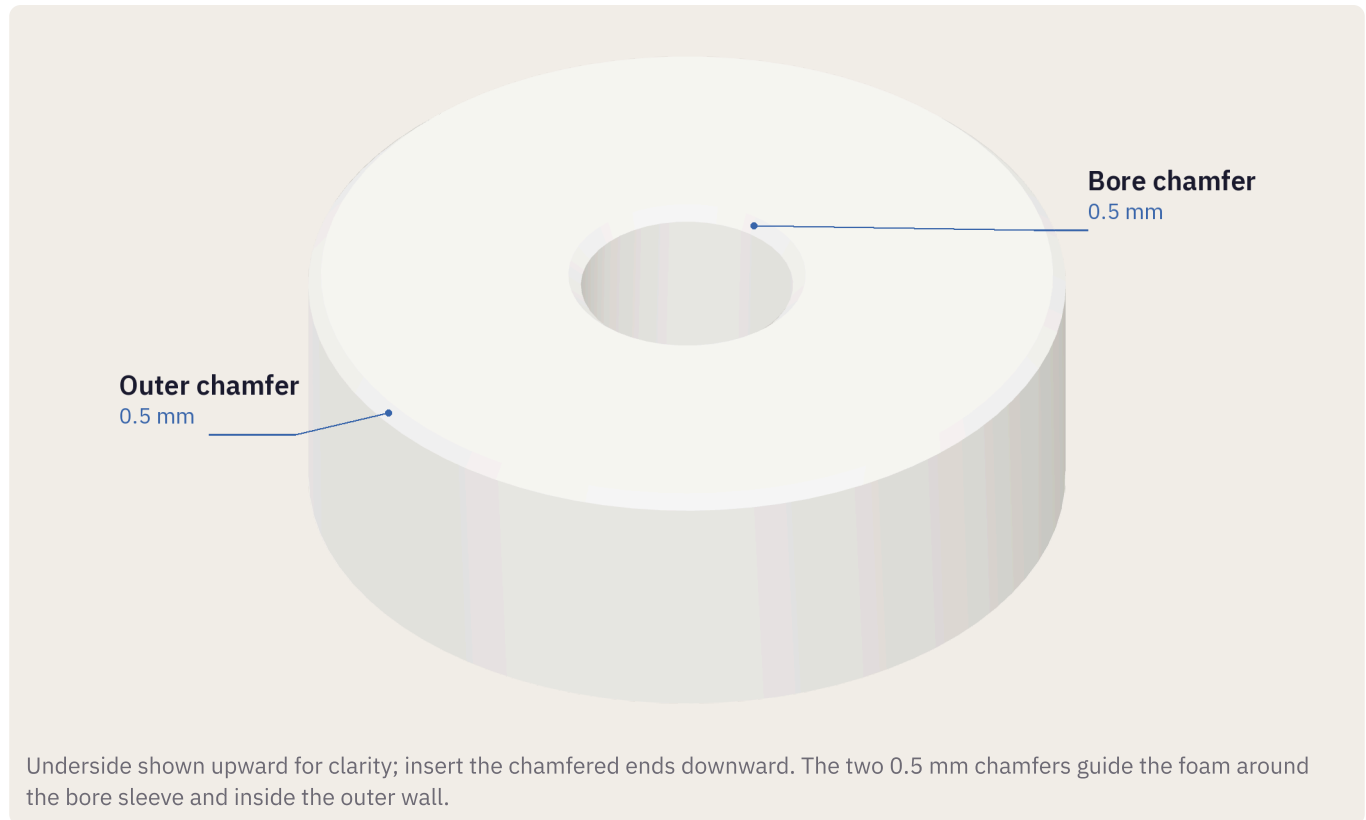


- 1 After the core finishes, leave the enclosure closed for the initial cooldown. Remove the plate once the bed reaches **35 °C or below**.
- 2 Gently flex the sheet to release the core. Set it upright on a clean surface.
- 3 Renew the thin glue coat in the print area and reinstall the Engineering plate.
- 4 In the same Aero project, select **plate 2**. Print with the same right nozzle and Aero settings.
- 5 Cool to **35 °C or below** and release the insert the same way.

Printing: about 37 min

Clean the entry edges

Both pieces have a chamfer on the lower outside edge and the lower bore edge. These ends enter the shell first.



- 1 Clip the thin brim from each cooled foam piece. Work around the edge with flush cutters.
- 2 Pare away the remaining brim lip with a sharp craft knife, following the printed edge. Leave the cylindrical fitting surfaces and the chamfers intact.
- 3 Clip loose strings from the central bores and the core's magnet pocket. Brush away all loose crumbs.
- 4 Keep the pieces dry and covered on the bench until the shell is ready.

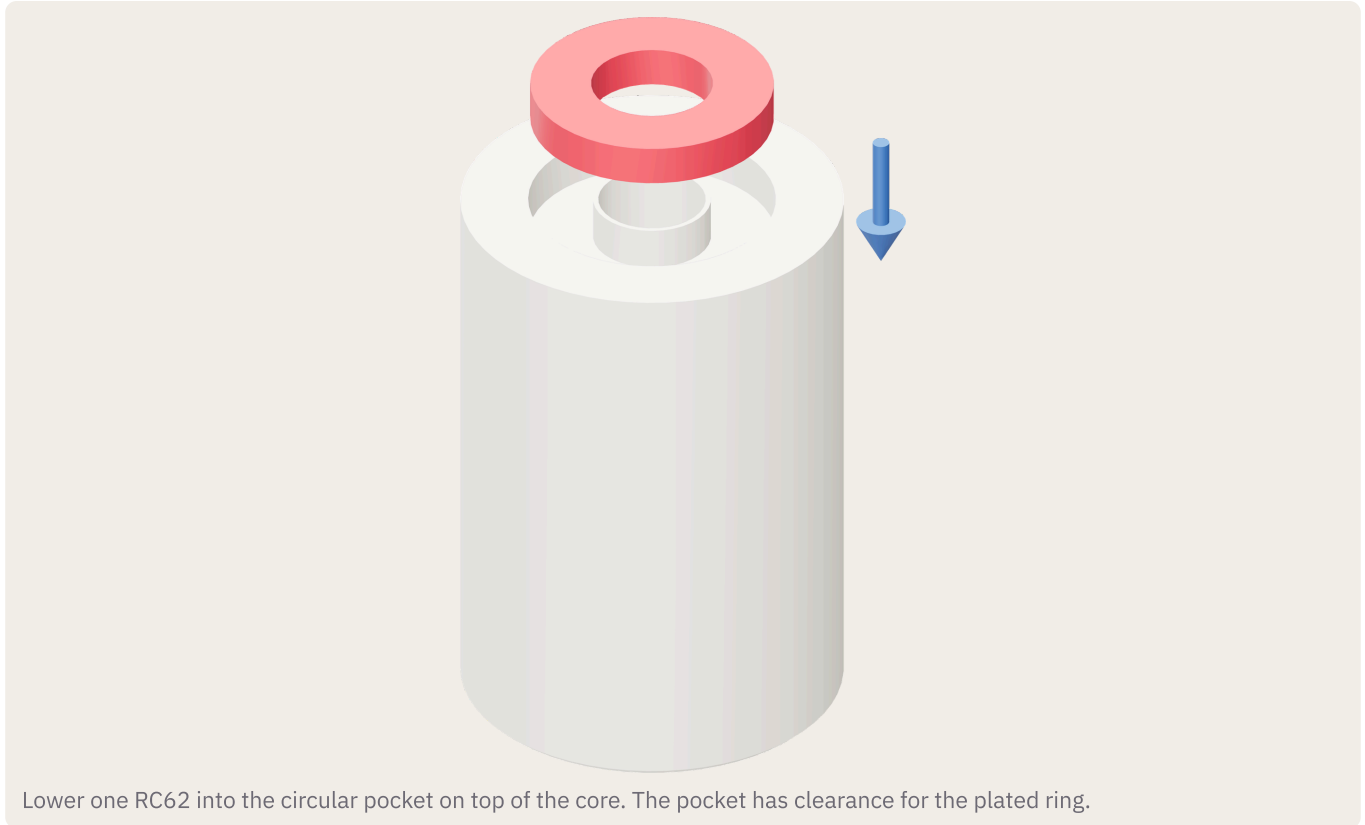
ORIENTATION

Tall core: magnet pocket up, chamfered end down.

Short insert: chamfered end down, broad flat face up.

Put the magnet in now

Prepare the core before starting PETG. At the printer, the core and magnet will go in together.



Lower one RC62 into the circular pocket on top of the core. The pocket has clearance for the plated ring.

- 1 Stand the core on its flat bottom, pocket upward.
- 2 Separate **one RC62** from its stack. Hold it by the edges and lower it squarely into the pocket.
- 3 Seat the ring flat against the pocket floor with light fingertip pressure. Either flat face can point upward for the reed-switch application.
- 4 Keep the prepared core upright beside the printer. Set the short insert beside it, chamfers downward.

CAPTURED BY THE INSERT

Use the ring as supplied. The printed pocket and upper insert hold it inside the finished float; assembly requires no adhesive.

Change to PETG

Let the warm Aero chamber cool, then fit the clean Textured PEI plate.

Before starting

≤35 °C

Chamber temperature after the Aero jobs.

During PETG

70 °C

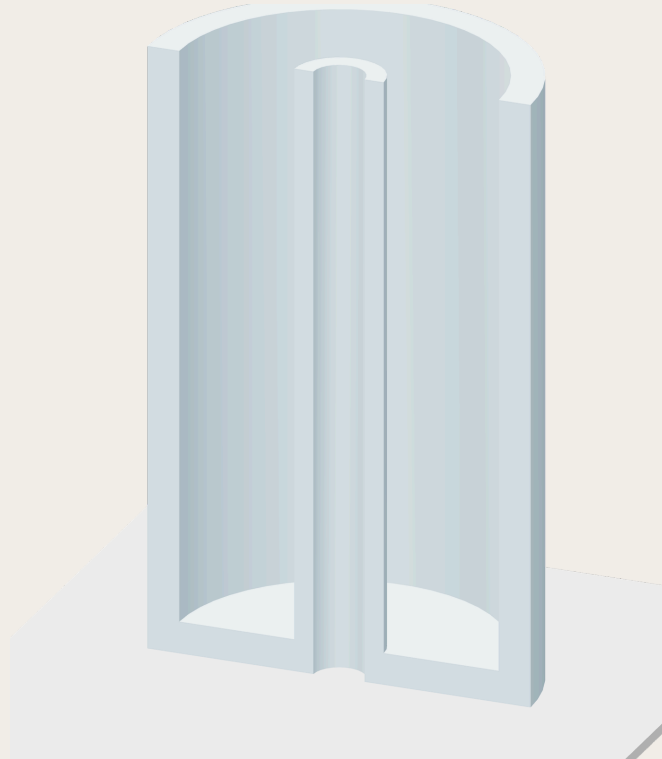
Textured PEI bed. Chamber heater off.

- 1 Let the chamber cool to **35 °C or below**. Wash the Textured PEI sheet with dish detergent and warm water, rinse and dry.
- 2 Install the Textured PEI sheet, holding it by the edges. Use its clean texture directly for this PETG job.
- 3 Open [magnetic-float.3mf](#) as a project. Keep the project's supplied **PETG Translucent settings** and map its material to the dried PETG Translucent in the left AMS feed.
- 4 Leave the prepared core, short insert and plastic scraper beside the printer. Read the insertion pages before starting.

Left nozzle	0.6 mm standard flow
Nozzle, first / remaining	255 / 260 °C
Flow / layer	1.02 / 0.18 mm
First layer	0.30 mm
Ordinary part fan	10-20%; auxiliary fan off

Print to the built-in pause

Start the single PETG plate. The shell stays fixed to the bed throughout insertion and roof printing.



Cutaway view. At the pause, the outer wall and central bore sleeve both end at 57 mm. The open annular space receives the foam.

- 1 Print **plate 1** using the left PETG feed. The job builds the 3 mm floor and both upright walls.
- 2 When the programmed pause occurs, let the toolhead finish parking. Open the door to reach the shell.
- 3 Leave the bed at **70 °C** and the sheet mounted. Keep the shell and the printer axes in place.

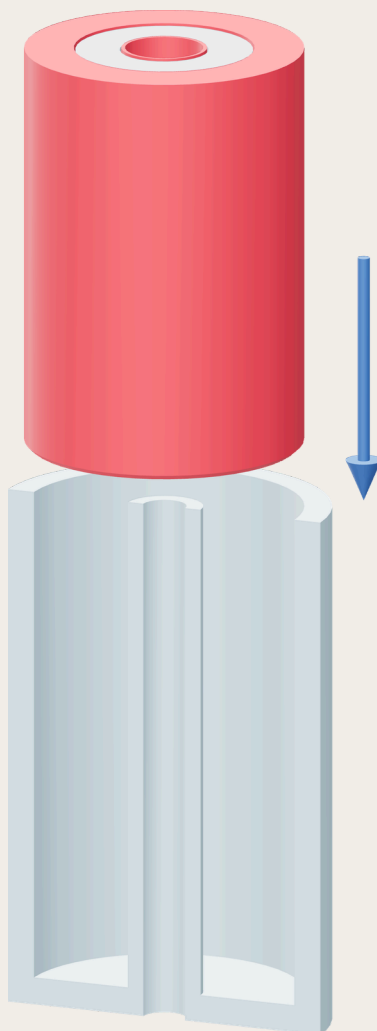
THE NEXT PRINTED LAYER

The pause is after **Z57.00 mm**, before the first roof layer at **Z57.18 mm**. Insert the prepared core and upper insert, then resume the same job.

Whole PETG job: about 2 h 02 min, plus your insertion time

Lower the prepared core

Hold it upright with the magnet at the top. The centre hole slides over the PETG bore sleeve.

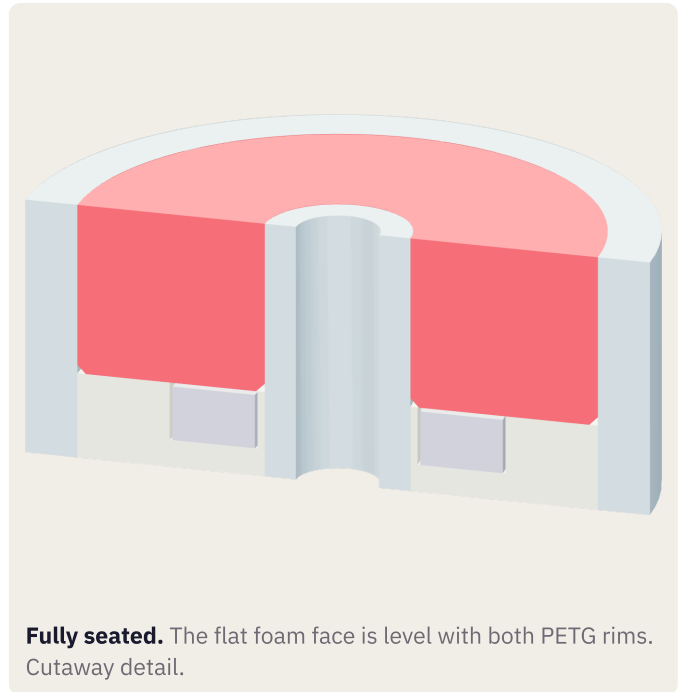
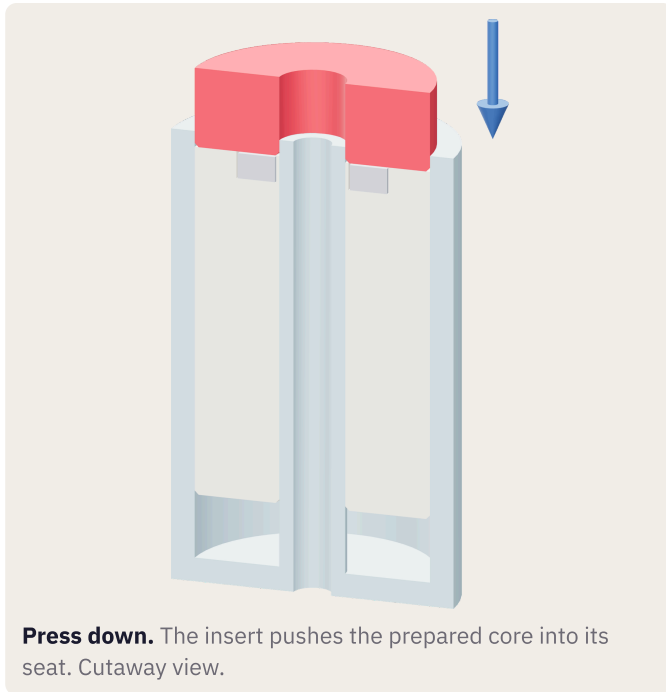


Cutaway view. Core and seated magnet enter as one prepared assembly. The lower chamfers guide both circular fits.

- 1 Centre the core over the sleeve. Lower it vertically, chamfered end first.
- 2 Press evenly around its outer top face to start it into the shell. Keep it square as it descends.
- 3 Keep the magnet seated in its pocket. Use the short insert for the final push on the next page.

Press the insert flush

The short insert pushes the core home, captures the magnet and supplies a flat surface for the roof.



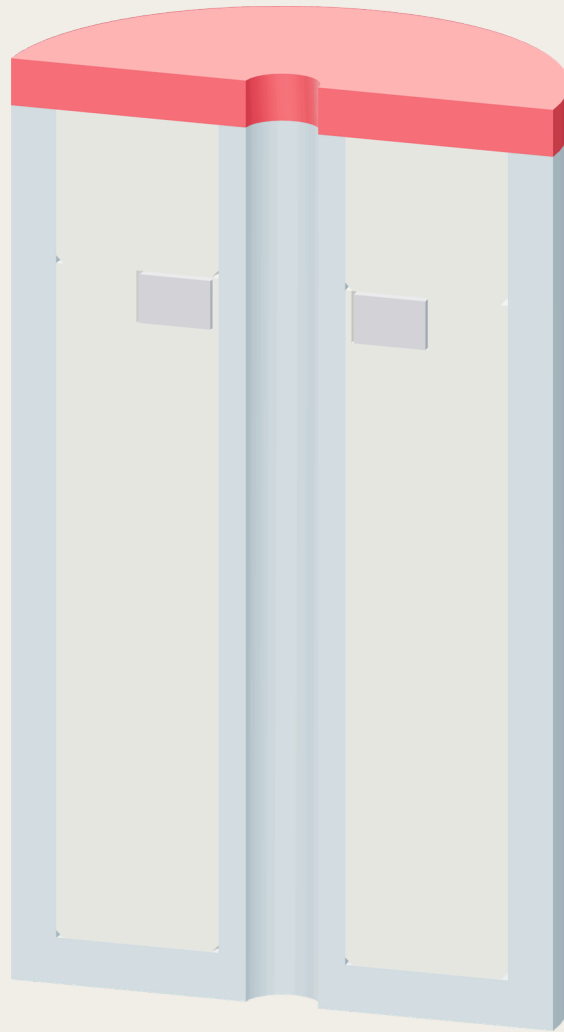
- 1 Lower the short insert over the bore sleeve with its **chamfered end down**. Rest it on the core.
- 2 Press downward evenly on opposite sides. Use the flat face of the clean plastic scraper to spread the last push across the insert.
- 3 Seat the insert with its flat upper face **level with the PETG outer rim and bore rim**. The core rests on the shell floor beneath it.
- 4 Lift the scraper clear. Brush or lift away any loose string from the top surface.

THE SEATING SURFACE

The roof prints across the whole foam annulus. Leave the upper face flat and level with the two PETG rims, and keep the central guide bore open.

Close the door and resume

The remaining 17 PETG layers make the 3.06 mm roof and enclose the complete assembly.



Cutaway view. The coral roof joins the outer wall to the bore sleeve over the seated foam. About 15 minutes of printing remain after the pause.

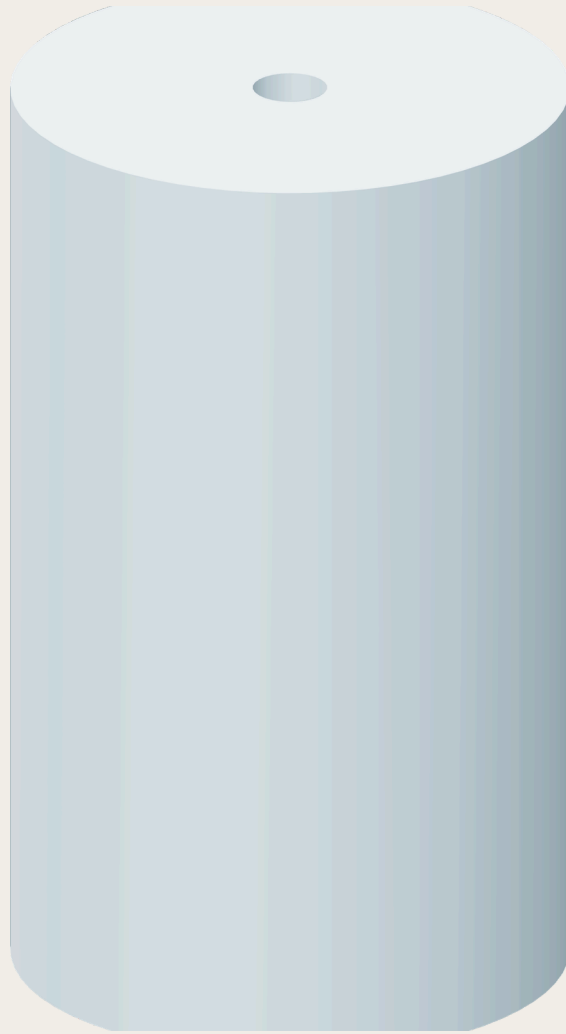
- 1 Remove your hands and the scraper from the printer.
- 2 Close the enclosure and press **Resume**. Let the printer restore its nozzle temperature and finish the roof.
- 3 Leave the float on the sheet until the bed reaches **35 °C or below**.

ONE CONTINUOUS SHELL

The final roof is part of the PETG print. There is no separate lid to glue on and no adhesive cure after printing.

Release and finish

The cooled print is the completed float. Keep the roof uppermost when fitting it to its guide.



Finished PETG float. The open centre bore passes through the body; the magnet remains enclosed near the top.

- 1 Lift the cooled Textured PEI sheet out of the printer. Flex it gently to release the float.
- 2 Clip off the outer brim and pare its remaining lip flush with the bottom edge.
- 3 Clip loose strings at the bore entrances and brush the exterior clean. Keep the printed PETG bore and shell walls intact.
- 4 Slide the **3.175 mm guide rod** through the **4.8 mm central bore**, with the printed roof facing upward.

TOP AND BOTTOM

The roof is the smooth, printed top surface above the magnet. The bottom carries the Textured PEI imprint. Keep that orientation when installing the float.

The build at a glance

Dry both spools overnight, finish the white pieces, then make the clear PETG shell around them.

ASA Aero drying	AMS HT · 80 °C × 8 h
PETG Translucent drying	AMS 2 Pro · 65 °C × 12 h
Aero project	<u>magnetic-float-aero.3mf</u>
Aero nozzle / bed / chamber	Right 0.4 mm · 270 / 90 / 60 °C
Aero plate	Engineering plate, thin Bambu liquid glue
Aero print order	Plate 1 core: 1 h 37 min Plate 2 insert: 37 min
Before PETG	Cool pieces and chamber to ≤35 °C. Trim both pieces and seat the RC62 in the core.
PETG project	<u>magnetic-float.3mf</u>
PETG nozzle / bed	Left 0.6 mm · 255/260 °C nozzle, 70 °C bed
PETG plate / chamber	Clean Textured PEI · chamber heater off
PETG printing time	2 h 02 min, plus insertion
Pause	After 57.00 mm, before 57.18 mm
Final closure	17 roof layers · 3.06 mm
Release	Bed ≤35 °C; flex sheet, remove brim

1 · Core

Magnet already seated.
Chamfers down.

2 · Insert

Chamfers down.
Press level with both rims.

3 · Resume

Tools clear.
Close enclosure.

Files and recipe sources

Use the supplied projects for this geometry. Their nozzle assignments, fit compensation and insertion pause are included.

Print projects

Foam parts	magnetic-float-aero.3mf
PETG shell	magnetic-float.3mf

The projects sit together under `hardware/printed-parts/cold-core/magnetic-float/` in this repository.

Recipe basis

Drying cycles	Bambu drying table: AMS HT / ASA Aero and AMS 2 Pro / PETG.
ASA Aero process	Bambu ASA Aero guide and H2C filament preset.
Clear PETG material	Bambu PETG Translucent, Clear 32101; same stock as the flavor reservoirs.
Engineering plate	Cleaning and use of Bambu liquid glue.
Textured PEI plate	Plate preparation and removal at 35 °C or below.
RC62 ring	Magnet dimensions, coating and material specification.

The shell uses Bambu PETG Translucent Clear (32101), with sealing settings derived from the water-holding reservoir recipe and adapted to this float. Aero uses the Bambu filament preset with solid foam fill. The 35 °C Aero handling/chamber changeover point is the selected shop procedure.

Prepared 1 October 2026. Illustrations are drawn from the float CAD. Clear PETG is shaded pale blue for clarity; warm white is Aero, and coral marks the part being added. Cutaway faces and insertion arrows explain assembly; they are not extra parts.